NARAYANA COACHING CENTER
NARAYANA IIT-JEE / NEET / FOUNDATIONS
44 YEARS OF EXCELLENCE

XI_IIT_IC_IR Date: 12-02-2024

Time: 3:00 hours **CTA – 05** Max. Marks:180

JEE – ADVANCE – 2022 – P1 – Model

IMPORTANT INSTRUCTIONS

PHYSICS

Section	Question Type	+Ve Marks	- Ve Marks	No. of Qs	Total marks
Sec – I (Q.N : 1 –8)	Questions with Numerical Answer Type	3	0	8	24
Sec – II (Q.N : 9 –14)	Questions with One or more Answer Type	4	-2	6	24
Sec – III (Q.N : 15 – 18)	Questions with Matrix Matching	3	1	4	12
Total				**18**	**60**

CHEMISTRY

Section	Question Type	+Ve Marks	- Ve Marks	No. of Qs	Total marks
Sec – I (Q.N : 19 –26)	Questions with Numerical Answer Type	3	0	8	24
Sec – II (Q.N : 27 –32)	Questions with One or more Answer Type	4	-2	6	24
Sec – III (Q.N : 33 – 36)	Questions with Matrix Matching	3	1	4	12
Total				**18**	**60**

MATHEMATICS

Section	Question Type	+Ve Marks	- Ve Marks	No. of Qs	Total marks
Sec – I (Q.N : 37 –44)	Questions with Numerical Answer Type	3	0	8	24
Sec – II (Q.N : 45 –50)	Questions with One or more Answer Type	4	-2	6	24
Sec – III (Q.N : 51 – 54)	Questions with Matrix Matching	3	1	4	12
Total				**18**	**60**

Exam Syllabus:

PHYSICS	**80%:** Thermal Expansion, Calorimetry, Heat Transfer, Thermal conduction, Theramal convection, and Radiation, Assumptions of KTG, Derivation of pressure, Heat Capapcities and specific hear capacities of solids and fluids, Mean free path, Maxwell's distribution Curve, First law of Thermodynamics, Work done by gas, state and process Indicator diagram, Cyclic process, molar heat capacity, Second law of thermodynamics, Heat engine and refrigerator, Kinematic Equations of SHM, Variation of velocity and acceleration during SHM, Dynamics of SHM, Energy of SHM, Physical Pendulum. **20%:** Unit and Dimension Complete chapter, Motion in straight line Complete chapter, Motion in a plane Complete chapter, Laws of Motion Complete chapter, Work power and Energy Complete chapter, System of Particles & Rotational Motion Complete Chapter, Mechanical properties of Fluids Complete Chapter, Mechanical properties of Solids Complete Chapter
CHEMISTRY	**(80%) :** p-block: 14th group elements General Organic Chemistry Alkane: preparation from alkene, alkyne, alkyl halides and carboxylic acids, Reaction of alkanes: halogenation (excluding steriochemistry) combustion, controlled oxidation, isomerisation, aromatization and pyrolysis. **(20%) :** Some Basic concept in Chemistry (Complete chapter); Atomic Structure (Complete chapter); Periodic Properties (Complete chapter); Chemical bonding and molecular structure (Complete chapter) ; State of Matter (Complete chapter); Thermodynamics (Complete chapter); Equilibrium (Complete chapter); Redox Reactions & (Volumetric Analysis) (Complete chapter); Hydrogen (Complete chapter); S-block element (Complete chapter); Group 13 Elements
MATHS	**80%** ELLIPSE: Definition, basics of ellipse, eccentric angle, parametric equation, position of point. Condition for tangency, equation of tangent, normal, chord of contact, pair of tangents, properties of ellipse, locus problems, Normal equation of chord having mid-point (x1, y1), pair of tangents, conjugate diameters chord of contact HYPERBOLA: Definition, basics of hyperbola, eccentric angle, parametric equation, position of point, Conjugate hyperbola, condition for tangency, equation of tangent, Normal, chord of contact, pair of tangents. Asymptotes, Rectangular hyperbola, Parametric form, Tangent and Normal of the Rectemgular hyperbola. PROPERTIES OF TRANGLES: Height and Distance, Properties of triangle, sine rule and cosine rule. Napier's analogy, projection formulae, Area of triangle in different form. Half angle formula, Circumcentre, incentre, Ortho centre, Centroid, Escribed circle, Regular polygon. **20%** Sequence and Series Complete chapter Trigonometric Equations Complete chapter Quadratic Equations Complete Chapter Theory of Equations Complete Chapter Binomial Theorem Complete Chapter Straight Line Complete chapter Circles complete chapter Complex numbers: complete chapter Permutation and Combination Complete chapter Parabola Complete Chapter

Rough Work

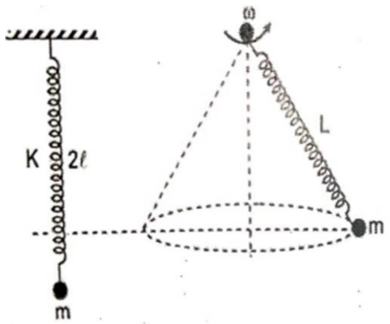
PHYSICS **Max. Marks : 60**

SECTION – I
NUMERICAL ANSWER TYPE
(Maximum Marks: 24)

This section contains **SIX (08)** questions. The answer to each question is a **NUMERICAL VALUE.**
For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, truncate/round-off the value to TWO decimal places.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If ONLY the correct numerical value is entered;
Zero Marks : 0 In all other cases.

1. 5.6 litres of helium gas at STP is adiabatically compressed to 0.7 litres. Taking the initial temperature to be T_1, the work done in the process is $-\frac{m}{n}RT_1$. Calculate m × n.
2. A gas undergoes a change of state during which 100 J of heat is supplied to it and it does 20 J of work. The system is brought back to its original state through a process during which 20 J of heat is rejected by the gas. Find the work done by the gas in the second process in joules.

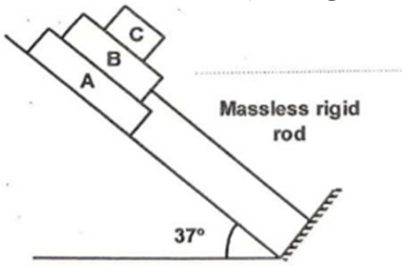
3. When a particle of mass m is suspended from a massless spring of natural length l, the length of the spring becomes $2l$. When the same mass moves in conical pendulum as shown in figure, the length of the spring becomes L. The radius of the circle of conical pendulum is $\left(\frac{L}{L-l}\right)\sqrt{L(L-nl)}$. Find the value of n.



4. A solid sphere of density 'ρ', specific heat capacity 'C' and radius 'R' is at an initial temperature 200 K. On allowing it to cool under the process of radiation then the time taken for it to cool to 100 K is found to be $\frac{7R\rho C}{8K\sigma}$, micro second. If 'σ' is Stefan's constant, then find the value of 'K'. Assume the surrounding temperature to be extremely small and emissivity of the outer surface of the sphere to be unity.

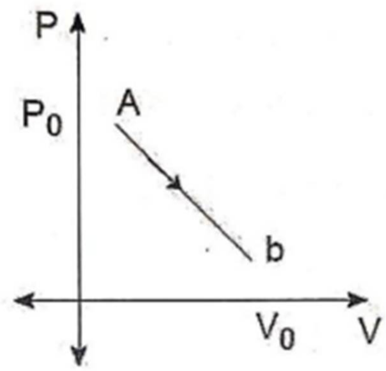
Rough Work

5. In the shown figure m_A, m_B and m_C are the masses of three blocks, the incline is frictionless m_A = 5 kg, m_B = 10 kg, m_C =2kg, coefficient of friction between A and B is 0.5 and between B and C is 0.1. The frictional force on block A is 6K Newton. Find the value of K. (Take g = 10 m/s^2).



6. A container filled with air under pressure P_o contains a soap bubble of radius R. The air pressure has been reduced to half isothermally and the new radius of the bubble becomes $\frac{5R}{4}$. If the surface tension of the soap water solution is S, P_o is found to be $\frac{16nS}{R}$ SI unit. Find n.

7. One mole of monoatomic gas undergoes a linear process A → B as shown in P-V diagram. Volume of gas from where process turn from an endothermic to an exothermic is $\frac{n}{(2n-2)}V_o$. Find n.



8. A 0.1 kg mass is suspended from a wire of negligible mass. The length of the wire is 1 m and its cross-sectional area is 4.9×10^{-7} m^2. If the mass is pulled a little in the vertically downward direction and released, it performs simple harmonic motion of angular frequency 140 rads^{-1}. If the Young's modulus of the material of the wire is $n \times 10^9$ Nm^{-2}, the value of n is______.

Rough Work

SECTION – II
(ONE MORE THAN QUESTIONS)
(MAXIMUM MARKS: 24)

This section contains **EIGHT (06)** questions.
Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer (s).
For each question, choose the option (s) corresponding to (all) the correct answer(s).
Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4 If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3 If all the four options are correct but ONLY three options are chosen
Partial Marks	:	+2 If three or more options are correct but ONLY two options are chosen and both of which are correct
Partial Marks	:	+1 If two or more options are correct but ONLY one option is chosen and it is a correct option:
Zero Marks	:	0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1 In all other cases.

9. A constant force F is applied on a spring block system as shown in figure. The mass of the block is M and spring constant is k. The block is placed over a smooth surface. Initially the spring was unstretched then,

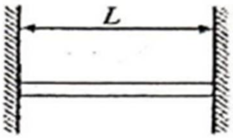
A) The block will execute SHM

B) Time period of oscillation is $2\pi\sqrt{\dfrac{2M}{k}}$

C) Time period of oscillation is $2\pi\sqrt{\dfrac{M}{k}}$

D) The speed of the block at distance x from initial position is $\sqrt{\dfrac{2Fx-kx^2}{M}}$

10. A horizontal rod (length L, mass m, cross section A, coefficient of linear expansion (α) fits exactly (without tension) between two rigid vertical supports. Choose the incorrect option.



A) Since the length of rod cannot increases beyond L, on increasing temperature, the effect of thermal expansion is neutralized by compressive forces developed by the wall.

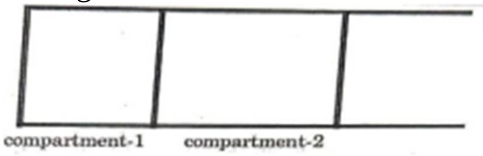
B) If the walls and rod are rough with mutual friction coefficient μ, then increment in temperature θ of the rod, required to prevent the rod from falling is $\Delta\theta = \dfrac{mg}{2\mu YA\alpha}$

C) If the walls and rod are rough with mutual friction coefficient μ, then increment in temperature θ of the rod, required to prevent the rod from falling is $\Delta\theta = \dfrac{mg}{4\mu YA\alpha}$

D) The stress developed in wire is $\alpha Y\Delta\theta$

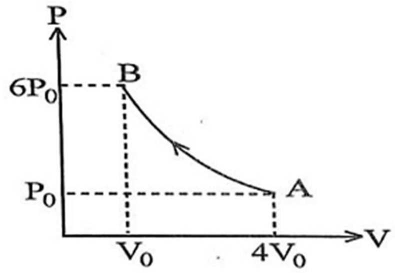
Rough Work

11. Figure shows a long fixed container which has two freely movable (without friction) pistons. The container and pistons are made of a thermally conducting material that allows very slow transfer to heat. First compartment of container is filled with 2 moles of an ideal monoatomic gas at 200 K and the 2nd compartment is filled with 1 mole of ideal diatomic gas at 500 K. Initially pressure of gases in both the compartments is same and equal to atmospheric pressure. Temperature of atmosphere is 300 K. Finally, gases achieve equilibrium. 'R' is the gas constant.



A) Heat transferred by gas in compartment-2 to its surrounding is 700 R
B) Heat transferred by gas in compartment-2 to the atmosphere is 500 R
C) Work done by gas in compartment-2 on the gas in compartment-1 is 200 R
D) Net heat transferred to gas in compartment-1 is 500 R

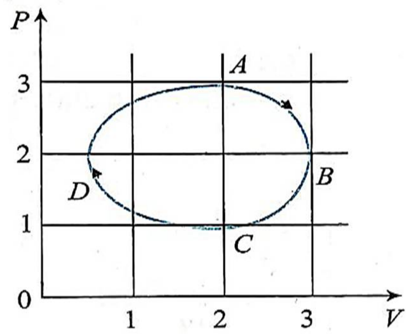
12. One mole of a diatomic gas is taken through a process A → B as shown in the figure. The gas obeys the relation $Q_{A\to B} + W_{A\to B} = 0$.



A) the molar heat capacity for process AB is 5R/4
B) the heat supplied in the process is $(5/2)P_oV_o$
C) for A to B, temperature initially decreases and then increases
D) the temperature goes on increasing from A to B

Rough Work

13. Figure shows the PV plot of an ideal gas taken through a cycle ABCDA. The part ABC is a semi-circle and CDA is half of an ellipse. Then,



A) the process during the path A → B is isothermal
B) heat flows out of the gas during the path B → C → D
C) work done during the path A → B → C is zero
D) Positive work is done by the gas in the cycle ABCDA

14. A container of fixed volume has a mixture of one mole of hydrogen and one mole of helium in equilibrium at temperature T. Assuming the gases are ideal, the correct statement(s) is(are):
A) The average energy per mole of the gas mixture is 2RT
B) The ratio of speed of sound in the gas mixture to that in helium gas is $\sqrt{6/5}$
C) The ratio of the rms speed of helium atoms to that of hydrogen molecules is 1/2
D) The ratio of the rms speed of helium atoms to that of hydrogen molecules is $1/\sqrt{2}$

Rough Work

SECTION – III
MATCHING ANSWER TYPE
(Maximum Marks: 12)

This section contains **TW0 (02)** List-Match sets.
Each List-Match set has TW0 (02) **Multiple Choice Questions**.
Each List-Match set has two lists: List-I and List-II.
List-I has Four entries (I), (II), (III) and (IV) and List-II has Six entries (P), (Q), (R), (S), (T) and (U).
FOUR options are given in each Multiple Choice Question based on List-I and List-II and ONLY ONE of these four options satisfies the condition asked in the Multiple Choice Question.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If ONLY the option corresponding to the correct combination is chosen;
Zero Marks : : O If none of the options is chosen (i.e., the question is unanswered);
Negative Marks : -1 In all other cases.

15. Consider a wire of length *l*, cross-sectional area A and Young's modulus Y and match the given two columns:

Column – I	Column – II
A) If the wire is pulled at its ends by equal and opposite forces of magnitude F so that it undergoes an elongation x, according to Hooke's law, F = Kx, K depends on	P) Young's modulus of elasticity (Y), of the wire
B) Let the wire be suspended vertically from a rigid support and a mass m is attached at its lower end, if the mass is slightly pulled and released, it executes SHM. Time period of oscillation depends on	Q) elongation (x) of the wire
C) If the thick inflexible wire is fixed between two rigid supports and its temperature is increased, thermal stress developed in the wire depends on	R) length of the wire (l)
D) If the wire is stretched from length l to $l + x$, then work done in stretching the wire depends on	S) Area of cross section of the wire (A)

A) A – P, S B – P, R, C – P, D – P, R, S
B) A – P, R, S, B – P, R, S, C – P, Q, R, S, D – P, Q, R, S
C) A – P, R, S, B – P, R, S, C – P, D – P, Q, R, S
D) A – Q, R, S, B – P, R, C – Q, R, S, D – P, R, S

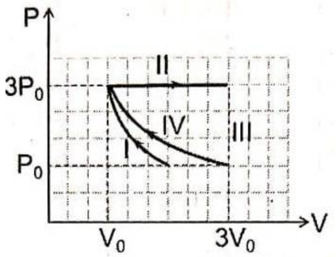
16. A particle of mass 2 kg is moving on a straight line under the action of force F = (8 – 2x) N. The particle is released from rest at x = 6 m. For the subsequent motion, match the following (all the values in column II are in their SI units):

Column I	Column II
i. Equilibrium position is at x.	a. $\pi/4$
ii. Amplitude of SHM is	b. $\pi/2$
iii. Time take to go directly from x = 2 to x = 4	c. 4
iv. Energy of SHM is	d. 6
v. Phase constant of SHM assuming equation of the form Asin(ωt + ϕ).	e. 2

A) i → c, ii → e, iii → b, iv → c, v → b
B) i → a, ii → e, iii → d, iv → c, v → b
C) i → c, ii → e, iii → d, iv → c, v → b
D) i → c, ii → a, iii → b, iv → d, v → a

Rough Work

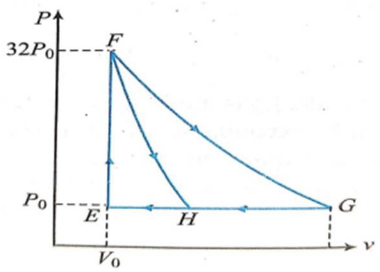
17. One mole of a monatomic ideal gas undergoes four thermodynamic processes as shown schematically in the PV-diagram below. Among these four processes, one is isobaric, one is isochoric, one is isothermal and one is adiabatic. Match the processes mentioned in Column -I with the corresponding statements in Column -II.



Column – I	Column – II
P: In process I	1. Work done by the gas is zero
Q: In process II	2. Temperature of the gas remains unchanged
R: In process III	3. No heat is exchanged between the gas and its surroundings
S: In process IV	4. Work done by the gas is 6 P_oV_o

A) P-4, Q-3, R-1, S-2
B) P-1, Q-3, R-2, S-4
C) P-3, Q-4, R-1, S-2
D) P-3, Q-4, R-2, S-1

18. One mole of mono-atomic ideal gas is taken along two cyclic processes E → F → G → E and E → F → H → E as shown in the PV diagram. The process involved are purely isochoric, isobaric, isothermal or adiabatic.



Match the paths in Column I with the magnitudes of the work done in Column II and select the correct answer using the codes given below the lists:

Column I	Column II
i. G → E	a. $160P_oV_o \ln 2$
ii. G → H	b. $36P_oV_o$
iii. F → H	c. $24P_oV_o$
iv. F → G	d. $31P_oV_o$

A) i-d, ii-c, iii-b, iv-a B) i-d, ii-c, iii-a, iv-b C) i-c, ii-a, iii-b, iv-d D) i-a, ii-c, iii-b, iv-d

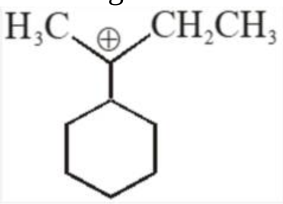
Rough Work

CHEMISTRY **Max. Marks: 60**

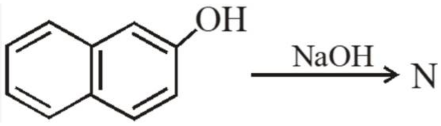
SECTION – I
(NUMERICAL ANSWER TYPE)

• This section contains **EIGHT (08)** questions.
• The answer to each question is a **NUMERICAL VALUE.**
• For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/roundoff** the value to **TWO** decimal places.
• Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +3 **ONLY** if the correct numerical value is entered.
Zero Marks: 0 In all other cases.

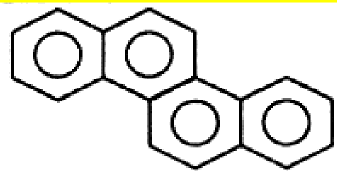
19. The total number of contributing structures showing hyperconjugation (involving C-H bonds) for the following carbocation is.



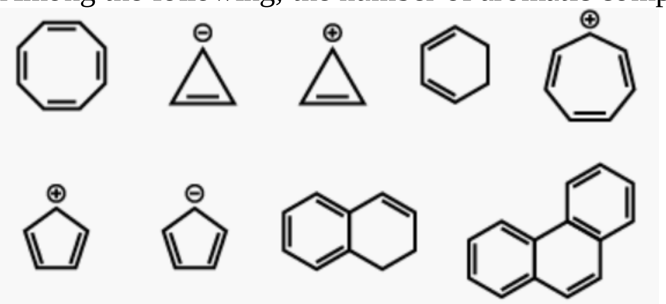
20. The number of resonance structures for N is:



21. Total Number of DBE value in:



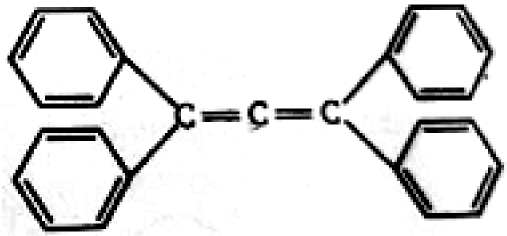
22. Among the following, the number of aromatic compound(s) is



23. Three moles of B_2H_6 are completely reacted with methanol. The number of boron containing product formed is____________________.

Rough Work

24. How many 2° Carbon atoms in the following?



25. Among the following, Number of Correct Statements are

A. $Al(CH_3)_3$ has the three-centre two-electron bonds in its dimeric structure.

B. Lewis acidity of BCl_3 is greater than that of $AlCl_3$

C. BH_3 has three-centre two-electron bonds in its dimeric structure.

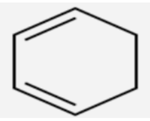
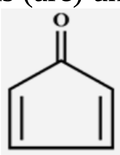
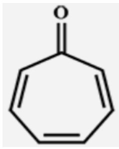
D. $AlCl_3$ has three-centre two-electron bonds in its dimeric structure.

26. What is the number of geometrical isomers formed with molecular formula C_2 (Cl)(Br) (I) (F)

SECTION – II
(ONE OR MORE ANSWER TYPE)

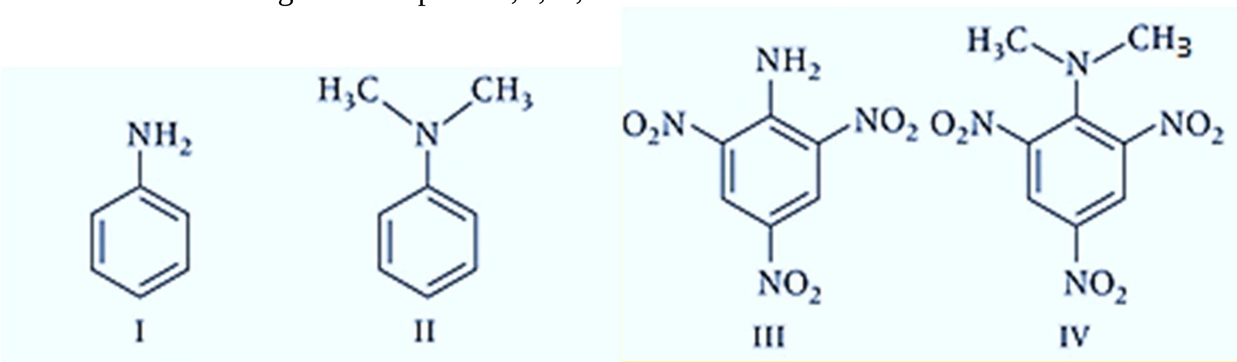
- This section contains **SIX (06)** questions.
- Each question has FOUR options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:
  Full Marks : +4 ONLY if (all) the correct option(s) is(are) chosen;
  Partial Marks : +3 If all the four options are correct but ONLY three options are chosen;
  Partial Marks : +2 If three or more options are correct but ONLY two options are chosen, both of which are correct;
  Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;
  Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
  Negative Marks : -2 In all other cases.

27. Which of the following molecules, on pure from, is (are) unstable at room temperature?

A)  B)  C)  D) 

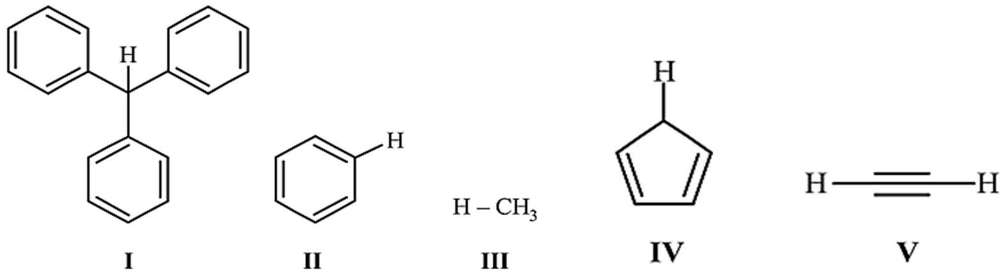
Rough Work

28. Consider the following four compounds, I, II, III and IV.



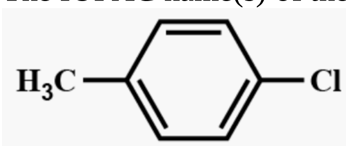
A) The order of basicity is II>I>III>IV>
B) The magnitude of pK_b difference between I and II is more than that between III and IV.
C) Resonance effects are greater in III than in IV.
D) Steric effects make compund IV more basic than III.

29. With respect to the compounds, I-V, choose the correct statements(s)



A) The acidity of compound I is due to delocalisation in the conjugate base.
B) The conjugate base of compound IV is aromatic.
C) Compound II becomes more acidic when it has a $-NO_2$ substituent.
D) The acidity of compounds follows the Order I>IV>V>II>III.

30. The IUPAC name(s) of the following compound is (are)



A) 1-chloro-4-methylbenzene B) 4-chlorotoluene
C) 4-methylchlorobenzene D) 1-methyl-4-chlorobenzene

31. Each of the following options contains a set of four molecules. Identify the options(s) where all four molecules posse's permanent dipole moment at room temperature.

A) SO_2, C_6H_5Cl,H_2Se,BrF_5 B) $BeCl_2$,CO_2,BCl_3,$CHCl_3$
C) NO_2, NH_3,$POCl_3$,CH_3Cl D) BF_3, O_3,SF_6,XeF_6

Rough Work

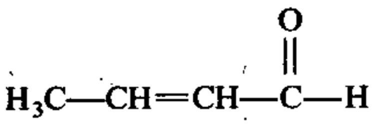
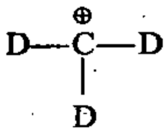
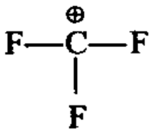
32. Aqueous solution of HNO_3, KOH and CH_3COOH and CH_3COONa of identical concentrations are provided. The pairs of solutions which form a buffer upon mixing is (are)

A) HNO_3 and CH_3CCOOH
B) KOH and CH_3COONa
C) HNO_3 and CH_3COONa
D) CH_3COOH and CH_3COONa

SECTION – III
(MATRIX MATCHING ANSWER TYPE)

- This section contains FOUR (04) Matching List Sets.
- Each set has ONE Multiple Choice Question.
- Each set has TWO lists: List-I and List-II.
- List-I has Four entries (I), (II), (III) and (IV) and List-II has Five entries (P), (Q), (R), (S) and (T).
- FOUR options are given in each Multiple-Choice Question based on List-I and List-II and ONLY ONE of these four options satisfies the condition asked in the Multiple-Choice Question.
- Answer to each question will be evaluated according to the following marking scheme:
  Full Marks : +3 ONLY if the option corresponding to the correct combination is chosen;
  Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
  Negative Marks: −1 In all other cases.

33. Match the Column

Column (I)	Column (II)
(a) 	*P.* Resonance
(b) 	*Q.* Hyperconjugation
(c) 	*R.* Inductive effect
(d) 	*S.* Non planar

A) (a) → P, Q, R (b)-R (c)-P, R (d)-S
B) (a) →P, R (b)-R, (c)-S, (d)-P
C) (a) → P, Q, R (b) -S (c) -P, R (d)-R
D) (a) →Q, S (b)-Q, (c)-P, (d)-R

Rough Work

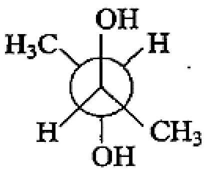
34. Match the Column.

Column (I) Compounds	**Column (II)** pK_a values
(a) $H_3C-\overset{O}{\overset{\parallel}{C}}-CH_2-\overset{O}{\overset{\parallel}{C}}-OEt$	P. 3.5
(b) $H_3C-\overset{O}{\overset{\parallel}{C}}-CH_2-\overset{O}{\overset{\parallel}{C}}-CH_3$	Q. 10.7
(c) $H-\overset{O}{\overset{\parallel}{C}}-CH_2-\overset{O}{\overset{\parallel}{C}}-H$	R. 8.9
(d) $H-\overset{O}{\overset{\parallel}{C}}-CH_2-\overset{O}{\overset{\parallel}{N}}\rightarrow O$	S. 0.4

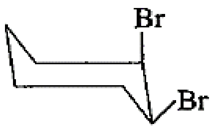
A) (a)-Q, (b)-R, (c)-P, (d)-S
B) (a)-P, (b)-Q, (c)-R, (d)-S
C) (a)-S, (b)-P, (c)-R, (d)-Q
D) (a)-Q, (b)-R, (c)-S, (d)-P

35. Match the Column.

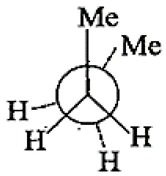
Column (I) | **Column (II)**

(a) 

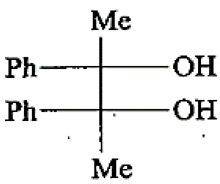
P. Meso

(b) 

Q. *Anti* conformer

(c) 

R. *Cis*-isomer

(d) 

S. Eclipsed conformers

A) (a)-S, P (b)-Q, R (c)-Q, S, (d)-R, P
B) (a)−P, Q; (b)− R (c)− S (d)−P, S
C) (a)−Q, R (b)− S, P (c)− Q, S (d)−R, P
D) (a)−P, Q (b)− P, R (c)− P, Q (d)−P, R

Rough Work

36. Match the transformation in column-I with appropriate options in column-II.

Column-I		Column-II	
a)	$CO_2(s) \to CO_2(g)$	P)	phase transition
b)	$CaCO_3(s) \to CaO(s) + CO_2(g)$	Q)	allotrapic change
c)	$2H. \to H_2(g)$	R)	ΔH *is positive*
d)	$P_{(white,solid)} \to P_{(red,solid)}$	S)	ΔS *is Positive*
		T)	ΔS is Negative

A) (a) → P, R, S; (b) → R, S; (c) → t; (d) → P, Q, T
B) (a) → R, S; (b) → P, Q, S; (c) → R; (d) → P, Q,T
C) (a) →R, S, T (b) →P, Q (c) →t, (d) →P, R
D) (a) →P, Q, (b) →R, s (c) →t, (d) →P, Q, T

Rough Work

MATHEMATICS **Max. Marks: 60**

SECTION – I
(NUMERICAL ANSWER TYPE)

- This section contains **EIGHT (08)** questions.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/roundoff** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
  Full Marks : +3 **ONLY** if the correct numerical value is entered;
  Zero Marks : 0 In all other cases.

37. If a tangent of slope 2 of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is normal to the circle $x^2+y^2+4x+1=0$, then the maximum value of ab is

38. If $\left(\sqrt{3}\right)bx+ay=2ab$ touches the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, the eccentric angle of the point of contact is $\frac{\pi}{\lambda}$, then value of λ is

39. If $x, y \in R$, satisfies the equation $\frac{(x-4)^2}{4}+\frac{y^2}{9}=1$, then the difference between the largest and the smallest value of the expression $\frac{x^2}{4}+\frac{y^2}{9}$ is.

40. The value of a for the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1,(a>b)$, if the extremities of the latus rectum of the ellipse having positive ordinates lie on the parabola $x^2=-2(y-2)$ is

41. If the variable line $y=kx+2h$ is tangent to an ellipse $2x^2+3y^2=6$, then the locus of P(h, k) is a conic C whose eccentricity is e. then the value of $3e^2$ is.

42. The length of the transvers axis of the rectangular hyperbola xy = 18 is

43. The eccentricity of the conjugate hyperbola of the hyperbola $x^2-3y^2=1$ is

44. The locus of the point of intersection of the lines $\sqrt{3}x-y-4\sqrt{3}t=0$ *and* $\sqrt{3}tx+ty-4\sqrt{3}=0$ (where t is a parameter) is a hyperbola whose eccentricity is.

Rough Work

SECTION – II
(ONE OR MORE ANSWER TYPE)

- This section contains **SIX (06)** questions.
- Each question has FOUR options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:
  Full Marks : +4 ONLY if (all) the correct option(s) is(are) chosen;
  Partial Marks : +3 If all the four options are correct but ONLY three options are chosen;
  Partial Marks : +2 If three or more options are correct but ONLY two options are chosen, both of which are correct;
  Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;
  Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
  Negative Marks : -2 In all other cases.

45. If the circle $x^2+y^2=a^2$ intersects the hyperbola xy = c^2 at four points $P(x_1,y_1), Q(x_2,y_2), R(x_3,y_3)$ *and* $S(x_4,y_4)$, then

A) $x_1+x_2+x_3+x_4=0$ B) $y_1+y_2+y_3+y_4=0$
C) $x_1x_2x_3x_4=c^4$ D) $y_1y_2y_3y_4=c^4$

46. The lines parallel to the normal to the curve xy = 1 is/ are

A) $3x+4y+5=0$ B) $3x-4y+5=0$
C) $4x+3y+5=0$ D) $3y-4x+5=0$

47. The sides of ΔABC satisfy the equation $2a^2+4b^2+c^2=4ab+2ac$. Then

A) the triangle is isosceles B) the triangle is obtuse
C) $B=\cos^{-1}(7/8)$ D) $A=\cos^{-1}(1/4)$

48. If the sines of the angle A and B of a triangle ABC satisfy the equation $c^2x^2-c(a+b)x+ab=0$, then the triangle

A) is acute angled B) is right angled
C) is obtuse angled D) satisfies the equation sin A + cos A= $\frac{(a+b)}{c}$

49. In triangle ABC if $2a^2b^2+2b^2c^2=a^4+b^4+c^4$, then angle B is equal to

A) 45° B) 135°
C) 120° D) 60°

50. If $\sin^3\theta+\sin\theta\cos\theta+\cos^3\theta=1$, then θ is equal to ($n\in Z$)

A) $2n\pi$ B) $2n\pi+\frac{\pi}{2}$
C) $2n\pi-\frac{\pi}{2}$ D) $n\pi$

Rough Work

SECTION – III
(MATRIX MATCHING ANSWER TYPE)

- This section contains FOUR (04) Matching List Sets.
- Each set has ONE Multiple Choice Question.
- Each set has TWO lists: List-I and List-II.
- List-I has Four entries (I), (II), (III) and (IV) and List-II has Five entries (P), (Q), (R), (S) and (T).
- FOUR options are given in each Multiple Choice Question based on List-I and List-II and ONLY ONE of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme:
  Full Marks : +3 ONLY if the option corresponding to the correct combination is chosen;
  Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
  Negative Marks : −1 In all other cases.

51. If α and β are roots of the equation $x^2-8x+4=0$, then match the following lists:

Column-I		Column-II	
	If $\alpha, a_1, a_2, a_3, a_4, a_5, \beta$ are in A.P., then $\sum_{i=1}^{5} a_i$ is divisible by		3
	If $\alpha, a_1, a_2, a_3, a_4, a_5, \beta$ are in G.P., then $\sum_{i=1}^{5} \log_2 a_i$ is divisible by		4
	If $\alpha^2, (a_1)^2, (a_2)^2, (a_3)^2, (a_4)^2, (a_5)^2, \beta^2$ are in A.P., then $\sum_{i=1}^{5} a_i^2$ is divisible by		5
	If $\alpha, a_1, a_2, a_3, a_4, a_5, \beta$ are in H.P., then $\left(\frac{1}{\sum_{i=1}^{5} \frac{1}{a_i}}+1\right)$ is divisible by		7
			6

Codes:

A) $a \to p, r;\ b \to q, p;\ c \to q, r;\ d \to s, t$

B) $a \to q, p;\ b \to p, s;\ c \to r, p;\ d \to r, t$

C) $a \to p, r;\ b \to r, s;\ c \to p, t;\ d \to q, p$

D) $a \to q, r;\ b \to r;\ c \to q, r, s;\ d \to p, t$

Rough Work

52. Match the following lists:

List-I (Number of positive integers of m for which)		List-II	
a)	One root is positive and the other is negative for the equation $(m-2)x^2-(8-2m)x-(8-3m)=0$	p)	0
b)	Exactly one root of equation $x^2-m(2x-8)-15=0$ lies in interval (0, 1)	q)	Infinite
c)	The equation $x^2+2(m+1)x+9m-5=0$ has both roots negative	r)	1
d)	the equation $x^2+2(m-1)x+m+5=0$ has both roots lying on either sides of 1	s)	2
		t)	3

Code:

A) $a\to p;\ b\to r;\ c\to q;\ d\to p$

B) $a\to q;\ b\to s;\ c\to p;\ d\to r$

C) $a\to r;\ b\to q;\ c\to p;\ d\to q$

D) $a\to s;\ b\to p;\ c\to q;\ d\to r$

53. Match the Column.

Column-I		Column-II	
a)	If lines $3x+y-4=0,\ x-2y-6=0$ and $\lambda x+4y+\lambda^2=0$ are concurrent then the value of λ is	p)	-4
b)	If the points $(\lambda+1,1),(2\lambda+1,3)$ *and* $(2\lambda+2,2\lambda)$ are collinear, then the value of λ is	q)	$-1/2$
c)	If the line $x+y-1-\|\lambda/2\|=0$, passing through the intersection of x – y + 1 = 0 and 3x + y – 5 = 0, is perpendicular to one of them, then the value of λ is	r)	4
d)	If the line y – x – 1 + λ = 0 is equidistant from the points (1, -2) and (3, 4), then λ is	s)	2
		t)	$1/2$

Codes:

A) $a\to p,r;\ b\to q,s;\ c\to p,s;\ d\to s$

B) $a\to p,s;\ b\to q,s;\ c\to p,r;\ d\to s$

C) $a\to p,r;\ b\to s\ \ c\to p,s;\ d\to q,s$

D) $a\to p,r;\ b\to s\ \ c\to q,s;\ d\to p,s$

Rough Work

54. Match the Column.

	Column-I		Column-II
a)	Points from which perpendicular tangents can be drawn to the parabola $y^2 = 4x$	p)	$(-1, 2)$
b)	Points from which only one normal can be drawn to the parabola $y^2 = 4x$	q)	$(3, 2)$
c)	Points at which chord $x - y - 1 = 0$ of the parabola $y^2 = 4x$ is bisected	r)	$(-1, -5)$
d)	Points from which tangents cannot be drawn to the parabola $y^2 = 4x$	s)	$(5, -2)$

Codes:

A) $a \to p, q;\ b \to r;\ c \to q, s;\ d \to q, r$

B) $a \to p, r;\ b \to p, r;\ c \to q;\ d \to q, s$

C) $a \to q, s;\ b \to r;\ c \to p, q;\ d \to q, r$

D) $a \to r;\ b \to q, r;\ c \to q, s;\ d \to p, q$

DETAILS OF PAPER SETTER :

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